

Equations over finite algebras

Erhard Aichinger

Institute for Algebra
Johannes Kepler University Linz
Linz, Austria

AAA105, Prague, June 2024

Supported by the Austrian Science Fund (FWF) : P33878

I. Systems of term equations

Systems of term equations

Let $\mathbf{A}$ be an algebra.

$\text{TERMSYSSAT}(\mathbf{A})$ is the following problem:

Given:

Terms $s_1(x_1, \dots, x_n), t_1(x_1, \dots, x_n), \dots, s_k(x_1, \dots, x_n), t_k(x_1, \dots, x_n)$.

Asked:

Is there $\mathbf{a} \in A^n$ with $s_1^{\mathbf{A}}(\mathbf{a}) = t_1^{\mathbf{A}}(\mathbf{a}), \dots, s_k^{\mathbf{A}}(\mathbf{a}) = t_k^{\mathbf{A}}(\mathbf{a})$?

Remarks:

- ▶ The answer is always **yes** if $\mathbf{A}$ has a one-element subuniverse: groups, lattices.
- ▶ Allowing constants yields $\text{POLSYSSAT}(\mathbf{A})$, which can be harder.

Computational complexity of $\text{TERMSYSSAT}(\mathbf{A})$

One can solve the equations by solving a **constraint satisfaction problem**.

Idea: (Larose, Zádori 2006)

Instead of solving

$$f(g(x_1, x_2)) = f(x_1),$$

solve

$$(x_1, x_2, y_1) \in g^\circ, (y_1, y_2) \in f^\circ, (x_1, y_2) \in f^\circ, \text{ where}$$

$$g^\circ = \{(a_1, a_2, b) \in A^3 \mid g(a_1, a_2) = b\}$$

is the **graph** of g .

This reduces $\text{TERMSYSSAT}(A; f, g)$ to $\text{CSP}(A; f^\circ, g^\circ)$.

Computational complexity of $\text{TERMSYSSAT}(\mathbf{A})$

For an algebra $\mathbf{A} = (A; F)$, let $\mathbf{A}^\circ := (A; \{f^\circ \mid f \in F\})$.

As a consequence of the Bulatov-Zhuk-Dichotomy (2017) (in the form of Barto, Krokhin, Willard (2017)), one obtains:

Theorem (cf. [Mayr, MFCS 2023]).

(Assume $\mathbf{P} \neq \mathbf{NP}$).

Let $\mathbf{A}$ be a finite algebra. Then $\text{TERMSYSSAT}(\mathbf{A}) \in \mathbf{P} \iff \mathbf{A}^\circ$ has a (not necessarily idempotent) Taylor polymorphism.

Otherwise $\text{TERMSYSSAT}(\mathbf{A})$ is $\mathbf{NP}$ -complete.

Computational complexity of $\text{TERMSYSSAT}(\mathbf{A})$

Question: Algebraic description when $\mathbf{A}^\circ$ has a (not necessarily idempotent) Taylor polymorphism.

Definition. Let $\mathbf{A}$ be a finite algebra.

$\text{Core}(\mathbf{A})$ is a minimal endomorphic image of $\mathbf{A}$ w.r.t $\subseteq$.

(Defined up to isomorphism)

Examples.

- ▶ $\mathbf{G}$ group. $\text{Core}(\mathbf{G}) = \{1\}$.
- ▶ $\mathbf{G}$ group. $\mathbf{G}^* := (G; *, {}^{-1}, (c_g)_{g \in G})$ its expansions with all constants from G .
Then $\text{Core}(\mathbf{G}^*) = G$.
- ▶ $\text{Core}((S_5; \circ, {}^{-1}, \underbrace{\text{id}, (1\ 2)}_{\text{nullary}})) = \{\text{id}, (1\ 2)\}$.

Computational complexity of $\text{TERMSYSSAT}(\mathbf{A})$

Theorem Larose, Zádori 2006

Let $\mathbf{A}$ be a finite algebra in a congruence modular variety. TFAE:

1. $\text{POLSYSSAT}(\mathbf{A}) = \text{TERMSYSSAT}(\mathbf{A}^*) \in \mathbf{P}$.
2. $\mathbf{A}$ is abelian.

Theorem Mayr 2023

Let $\mathbf{A}$ be a finite algebra in a congruence modular variety. TFAE:

1. $\text{TERMSYSSAT}(\mathbf{A}) \in \mathbf{P}$.
2. $\text{Core}(\mathbf{A})$ is abelian.

Both results also hold also if $1 \notin \text{typ}(V(\mathbf{A}))$ and $5 \notin \text{typ}(\{\mathbf{A}\})$.

TERMSYSSAT($\mathbf{A}$) vs. POLSYSSAT($\mathbf{A}$)

Theorem Mayr 2023.

Let $\mathbf{A}$ be a finite algebra of finite type. The following three problems are reducible to each other in **constant** time:

1. TERMSYSSAT($\mathbf{A}$).
2. TERMSYSSAT(Core($\mathbf{A}$)).
3. POLSYSSAT(Core($\mathbf{A}$)).

The meta-problem for systems of term equations

The meta-problem for TERMSYSSAT

Meta-problem for TERMSYSSAT (Assume $\mathbf{P} \neq \mathbf{NP}$)

Given: $\mathbf{A} = (A; f_1, \dots, f_k)$

Asked: Is $\text{TERMSYSSAT}(\mathbf{A}) \in \mathbf{P}$?

The meta-problem for TERMSYSSAT

Meta-problem for TERMSYSSAT (Assume $\mathbf{P} \neq \mathbf{NP}$)

Given: $\mathbf{A} = (A; f_1, \dots, f_k)$

Asked: Is $\text{TERMSYSSAT}(\mathbf{A}) \in \mathbf{P}$?

Asked: Does $\text{Core}(\mathbf{A}^\circ)$ have a Siggers polymorphism?

The meta-problem for TERMSYSSAT

Meta-problem for TERMSYSSAT (Assume $\mathbf{P} \neq \mathbf{NP}$)

Given: $\mathbf{A} = (A; f_1, \dots, f_k)$

Asked: Is $\text{TERMSYSSAT}(\mathbf{A}) \in \mathbf{P}$?

Asked: Does $\text{Core}(\mathbf{A}^\circ)$ have a Siggers polymorphism?

In cm varieties: **Asked:** Does $\mathbf{A}$ have an abelian core?

Theorem Mayr 2023

There is a **quasi-polynomial** algorithm that decides whether a given finite $\mathbf{A}$ in a cm variety has an abelian core.

$q(n)$ is **quasi-polynomial** if $\exists c, d, N > 0 \forall n \geq N : q(n) \leq c 2^{\log(n)^d}$.

Solving CSPs by solving term equations

Solving CSP's through systems of equations

Theorem.

For every finite relational structure $\mathbb{D}$ of finite type, there is a finite algebra $\mathbf{A}(\mathbb{D})$ such that $\text{CSP}(\mathbb{D})$ and $\text{TERMSYS SAT}(\mathbf{A}(\mathbb{D}))$ are polynomial time reducible to each other.

1. Klíma, Tesson, Thérien 2007:

Assume $\mathbb{D} = (D, \rho)$ is a digraph. $\mathbf{A}(\mathbb{D})$ is a semigroup with $5|D| + |\rho| + 1$ elements that satisfies $x^2 \approx x$ and $xyz \approx yxz$.

2. Broniek 2015:

Assume $\mathbb{D} = (D, R)$ with $R \subseteq D^r$. $\mathbf{A}(\mathbb{D})$ is a **unary** algebra with $|D| + |R| + 2$ elements and $r + 4$ unary operations.

Solving systems over modules

Solving systems over modules

Let $\mathbf{A}$ be an $\mathbf{R}$ -module.

- ▶ The polynomial algorithm provided by the theory uses the Bulatov-Dalmau-algorithm (2006) to solve instances of $\text{CSP}(\mathbf{A}^\circ)$, which has the Mal'cev term of $\mathbf{A}$ as a polymorphism.
- ▶ In practice, [Hermite-decomposition](#) is useful.

Solving systems over modules

We solve

$$\begin{pmatrix} 10 & 16 & 0 \\ 15 & 24 & 30 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 66 \end{pmatrix}$$

over $\mathbb{Z}$.

Solving systems over modules

We solve

$$\begin{pmatrix} 10 & 16 & 0 \\ 15 & 24 & 30 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 66 \end{pmatrix}$$

over $\mathbb{Z}$. To this end, we compute a $\mathbb{Z}$ -Basis of the row module of

$$\begin{pmatrix} -4 & -66 & 1 & 0 & 0 & 0 \\ 10 & 15 & 0 & 1 & 0 & 0 \\ 16 & 24 & 0 & 0 & 1 & 0 \\ 0 & 30 & 0 & 0 & 0 & 1 \end{pmatrix}$$

using the Hermite normal form (1851, polynomial time since 1979).

Solving systems over modules

We solve

$$\begin{pmatrix} 10 & 16 & 0 \\ 15 & 24 & 30 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 66 \end{pmatrix}$$

over $\mathbb{Z}$. We have

$$\text{row}\left(\begin{pmatrix} -4 & -66 & 1 & 0 & 0 & 0 \\ 10 & 15 & 0 & 1 & 0 & 0 \\ 16 & 24 & 0 & 0 & 1 & 0 \\ 0 & 30 & 0 & 0 & 0 & 1 \end{pmatrix}\right) = \text{row}\left(\begin{pmatrix} 2 & 3 & 0 & 5 & -3 & 0 \\ 0 & 30 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 2 & -1 & 2 \\ 0 & 0 & 0 & 8 & -5 & 0 \end{pmatrix}\right)$$

and thus $S = \{(2, -1, 2) + t(8, -5, 0) \mid t \in \mathbb{Z}\}$.

Solving systems over modules

Problem: Find all $(z_1, z_2, z_3) \in \mathbb{Z}[x, y]^3$ with

$$(10y)z_1 + 0z_2 + (4x)z_3 = 4x^3.$$

Solution: Compute the (reduced strong) Gröbner basis (with respect to a certain order) of the row module of

$$A' := \begin{pmatrix} -4x^3 & 1 & 0 & 0 & 0 \\ 10y & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 4x & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Solving systems over modules

We solve $(10y)z_1 + 0z_2 + (4x)z_3 = 4x^3$.

$$\text{row}\left(\begin{pmatrix} -4x^3 & 1 & 0 & 0 & 0 \\ 10y & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 4x & 0 & 0 & 0 & 1 \end{pmatrix}\right) = \text{row}\left(\begin{pmatrix} 2xy & 0 & x & 0 & -2y \\ 4x & 0 & 0 & 0 & 1 \\ 10y & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & x^2 \\ 0 & 0 & 2x & 0 & -5y \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}\right).$$

Hence $S = (0, 0, x^2) + \langle (2x, 0, -5y), (0, 1, 0) \rangle$.

II. One equation

Supernilpotent algebras

Theorem (Coordinatization, EA 2019).

Let $\mathbf{A}$ be supernilpotent of order p^α in a cm variety. Then there are operations $+, \times$ on A and $D \in \mathbb{N}$ such that

1. $\mathbb{F} = (A; +, \times)$ is a field,
2. for all $n \in \mathbb{N}$ and $p \in \text{Pol}_n(\mathbf{A})$, there is by $P \in \mathbb{F}[X_1, \dots, X_n]$ with $P^{\mathbb{F}} = p^{\mathbf{A}}$ and $\deg(f) \leq D$.

Corollary Kompatscher 2018.

Let $\mathbf{A}$ be supernilpotent in a cm variety. Then **one** equation

$$s(x_1, \dots, x_n) = t(x_1, \dots, x_n)$$

can be solved in polynomial time.

Supernilpotent algebras

Corollary Kompatscher 2018.

Let $\mathbf{A}$ be supernilpotent in a cm variety. Then **one** equation $s(x_1, \dots, x_n) = t(x_1, \dots, x_n)$ can be solved in polynomial time.

- ▶ Assume $|A| = p^\alpha =: q$.
- ▶ $\exists S, T \in \mathbb{F}_q[X_1, \dots, X_n] : S^{\mathbb{F}} = s^{\mathbf{A}}$ and $T^{\mathbb{F}} = t^{\mathbf{A}}$ with $\deg(S) \leq D$, $\deg(T) \leq D$.
- ▶ $\deg(1 - (S - T)^{q-1}) \leq (q - 1)D$.
- ▶ Let $(a_1, \dots, a_n)$ be the nonzero of $P := 1 - (S - T)^{q-1}$ with the smallest number of nonzero entries. WLOG $(a_1, \dots, a_n) = (a_1, \dots, a_k, 0, \dots, 0)$.
- ▶ $Q(X_1, \dots, X_k) := P(X_1, \dots, X_k, 0, \dots, 0)$ satisfies $\deg(Q) \geq k$.
- ▶ Hence $k \leq (q - 1)D$.
- ▶ Look for a solution of $s = t$ with at most $(q - 1)D$ entries different from $a \in A$.

Supernilpotent algebras

Why was it easy to solve equations?

- ▶ We could reduce the search space from $|A|^n$ to a **hitting set** of size $c(|A|) n^k$.
- ▶ There are either **no** or **many** solutions.

Theorem Warning 1935.

$$f \in \mathbb{F}_q[X_1, \dots, X_n].$$

If the polynomial f with $\deg(f) < n$ has a zero, then the number of zeros of f is at least $q^{n-\deg(f)}$.

Consequence: if there is a solution, picking $\mathbf{a}$ at random yields a solution with probability at least $q^{-\deg(f)} \rightsquigarrow$ linear time Monte Carlo-algorithm (Kawałek, Krzackowski 2020).

Solving equations over groups

Solving equations over groups

$\mathbf{G}$ a finite group. $\text{POLSAT}(\mathbf{G})$ asks whether

$$s(x_1, \dots, x_k, g_1, \dots, g_l) = t(x_1, \dots, x_k, g_1, \dots, g_l)$$

has a solution in G^k .

Input size: lengths of terms s and t .

- ▶ $\mathbf{G}$ nilpotent $\Rightarrow \text{POLSAT}(\mathbf{G}) \in \mathbf{P}$ (Horváth, 2011).
- ▶ $\mathbf{G}$ not solvable $\Rightarrow \text{POLSAT}(\mathbf{G})$ is **NP**-complete (Goldmann and Russell, 1999).
- ▶ $\mathbf{G}$ not solvable $\Rightarrow$ there exists $e \in \text{Pol}_1(\mathbf{A})$ such that $\text{POLSAT}(\mathbf{A} + e)$ has no subexponential algorithm, or the **exponential time hypothesis** fails (Rossi, EA 2024).
- ▶ $\mathbf{G} = P \rtimes A$ with P p -group and A abelian $\Rightarrow \text{POLSAT}(\mathbf{G}) \in \mathbf{P}$. (Földvári, Horváth 2019).

Even solving a fixed number of equations is in **P** (Nuspl 2021).

Solving equations over groups

Theorem Idziak, Kawałek, Krzaczkowski; Weiß 2020

If $\text{POLSAT}(S_4) \in \mathbf{P}$, then for every $\varepsilon > 0$, we can solve 3-SAT in time $O(2^{\varepsilon n})$, contradicting the [Exponential Time Hypothesis](#) by Impagliazzo and Paturi from 1999.

- ▶ The result is not just about S_4 , but about all groups of [Fitting length](#) (length of shortest composition series with nilpotent quotients) at least 3.
- ▶ An algorithm with running time $O(n^{c(\log(n))^d})$ is consistent with ETH, where n is the length of the input terms.

Solving equations over groups

- ▶ We make a (still unsuccessful) attempt to find an $O(n^{c(\log(n))^d})$ algorithm for $\text{POLSAT}(S_4)$.
- ▶ Easier problem: $\text{POLEQV}(S_4)$:
Input: p, q polynomial terms over S_4 .
Output: Is $\forall \mathbf{a} \in S_4^n : p(\mathbf{a}) = q(\mathbf{a})$ true?

Identity checking for groups

We want to check whether $p(x_1, \dots, x_n) = 1$ for all $\mathbf{x} \in S_4^n$.

Definition. $p \in \text{Pol}_k(S_4)$ is **absorbing** if for all $\mathbf{x}$: $1 \in \{x_1, \dots, x_k\} \Rightarrow p(x_1, \dots, x_k) = 1$.

- ▶ Suppose we can prove: **every nonconstant absorbing polynomial has length at least $2^{ck^{1/d}}$** .
- ▶ Then pick $\mathbf{a}$ with $p(\mathbf{a}) \neq 1$ with maximal amount of 1's. WLOG $\mathbf{a} = (a_1, \dots, a_k, 1, \dots, 1)$.
- ▶ Then $q(x_1, \dots, x_k) := p(x_1, \dots, x_k, 1, \dots, 1)$ is absorbing.
- ▶ Hence $\text{length}(p) \geq 2^{ck^{1/d}}$.
- ▶ Then $k \leq \frac{1}{c}(\log_2(\text{length}(p)))^d$.
- ▶ Input size m : greater than $\max(n, \text{length}(p))$.
- ▶ There are at most $m^{c_2 \log^d(m)}$ tuples with at most k entries $\neq 1 \rightsquigarrow$ **algorithm of complexity $O(m^{c \log^d(m)})$** .

Finite fields

Absorbing polynomial functions over finite fields

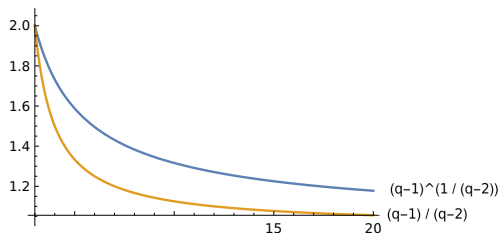
Definition. Let $\mathbb{F}$ be a field, $f \in \mathbb{F}[X_1, \dots, X_n]$ is **absorbing at a for S** : $\Leftrightarrow \forall (x_1, \dots, x_n) \in S^n : a \in \{x_1, \dots, x_n\} \Rightarrow f(x_1, \dots, x_n) = 0$.

Theorem (Grünbacher, Hametner, EA 2024).

$S \subseteq \mathbb{F}_q \setminus \{0\}$, $a \in S$. If $f \in \mathbb{F}_q[X_1, \dots, X_n]$ is absorbing at a for S and f is not identically 0 on S , then f contains at least $(\frac{q-1}{q-2})^n$ monomials.

Examples:

1. $\omega :=$ primitive element of $\mathbb{F}_4$,
 $S := \{1, \omega\}$. Then $f := \prod_{i=1}^n (X_i - \omega)$ is absorbing at ω for S and has 2^n monomials.
2. (Grünbacher) There is an absorbing function with at most $(\sqrt[q-2]{q-1})^n$ monomials (if $q-2 \mid n$).



Equation solving over finite fields

Intuition: If there is a solution, there is one in the neighborhood.

Theorem Grünbacher, Hametner, EA 2024

Let $q > 2$, $n \in \mathbb{N}$, $f \in \mathbb{F}_q[X_1, \dots, X_n] \setminus \{0\}$ with $M(f)$ monomials, $S \subseteq \mathbb{F}_q \setminus \{0\}$, $t := \frac{q-1}{q-2}$. Let

$$V(f) := \{\mathbf{x} \in S^n \mid f(\mathbf{x}) = 0\}.$$

If $V(f) \neq \emptyset$, then for every $\mathbf{a} \in S^n$, there is $\mathbf{b} \in V(f)$ with

$$d_H(\mathbf{a}, \mathbf{b}) \leq \frac{1}{\log_2(t)} (1 + (q-1) \log_2(M(f))).$$

This gives quasi-polynomial time algorithms for the following questions:

1. Does f have a zero in S^n ?
2. Does f vanish identically on S^n ?

III. Quasi-identities

Quasi-identities in universal algebra

- ▶ $\mathbf{A}$ algebra, s_i, t_i, u, v terms.
- ▶ We ask whether $S = \{\mathbf{x} \in A^n \mid \bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x})\}$ is contained in $U = \{\mathbf{x} \in A^n \mid u(\mathbf{x}) = v(\mathbf{x})\}$.
- ▶ This holds if the formula

$$\forall \mathbf{x} : \left(\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}) \right) \implies u(\mathbf{x}) = v(\mathbf{x})$$

holds in $\mathbf{A}$.

- ▶ Such a formula is called a **conditional identity** or **quasi-identity**.
- ▶ We want to determine the validity of this formula.

Quasi-identities in universal algebra

- ▶ $\mathbf{A}$ algebra, s_i, t_i, u, v terms.
- ▶ We ask whether $S = \{\mathbf{x} \in A^n \mid \bigwedge_{i \in \underline{k}} s_i^{\mathbf{A}}(\mathbf{x}) = t_i^{\mathbf{A}}(\mathbf{x})\}$ is contained in $U = \{\mathbf{x} \in A^n \mid u^{\mathbf{A}}(\mathbf{x}) = v^{\mathbf{A}}(\mathbf{x})\}$.
- ▶ This holds if the formula

$$\forall \mathbf{x} : \left(\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}) \right) \implies u(\mathbf{x}) = v(\mathbf{x})$$

holds in $\mathbf{A}$.

- ▶ Such a formula is called a **conditional identity** or **quasi-identity**.
- ▶ We want to determine the validity of this formula.

Algebras that satisfy the same quasi-identities

Some facts on quasi-identities:

- ▶ Classes of algebras defined by quasi-identities are called **quasivarieties**. The quasivariety generated by K is $ISP_u P_{\text{fin}} K$.
- ▶ Generalization: infinite pre-condition, finitely many variables:
Considered in **Universal Algebraic Geometry**. Closure operator: LSP .
- ▶ Generalization: infinite pre-condition, arbitrary many variables:
Closure operator ISP .
- ▶ $\mathbf{A}, \mathbf{B}$ finite, of finite type, $|\mathbf{A}| = n$. Then $\mathbf{A} \in Q(\mathbf{B}) \Leftrightarrow \mathbf{A} \in IS\{\mathbf{B}^{\binom{n}{2}}\}$.

Quasi-identity validity

Let $\mathbf{A}$ be an algebra. $\text{QUASIDVAL}(\mathbf{A})$ is the problem:

Given: A quasi-identity $\Phi := \forall \mathbf{x} : \left(\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}) \right) \implies u(\mathbf{x}) = v(\mathbf{x})$.

Here, s_i, t_i, u, v are **terms** in the language of $\mathbf{A}$ over the variables $\mathbf{x}$.

Asked: Does Φ hold in $\mathbf{A}$?

Quasi-identity validity

Let $\mathbf{A}$ be an algebra. $\text{QUASIDVAL}(\mathbf{A})$ is the problem:

Given: A quasi-identity $\Phi := \forall \mathbf{x} : \left(\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}) \right) \implies u(\mathbf{x}) = v(\mathbf{x})$.

Here, s_i, t_i, u, v are **terms** in the language of $\mathbf{A}$ over the variables $\mathbf{x}$.

Asked: Does Φ hold in $\mathbf{A}$?

Computational Complexity: For finite $\mathbf{A}$ of finite type, $\text{QUASIDVAL}(\mathbf{A})$ is in **co-NP**:

$\mathbf{a} \in A^n$ witnesses failure of Φ if $\left(\bigwedge_{i \in \underline{k}} s_i^{\mathbf{A}}(\mathbf{a}) = t_i^{\mathbf{A}}(\mathbf{a}) \right) \wedge u^{\mathbf{A}}(\mathbf{a}) \neq v^{\mathbf{A}}(\mathbf{a})$.

Quasi-identity validity

Let $\mathbf{A}$ be an algebra. $\text{QUASIDVAL}(\mathbf{A})$ is the problem:

Given: A quasi-identity $\Phi := \forall \mathbf{x} : \left(\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}) \right) \implies u(\mathbf{x}) = v(\mathbf{x})$.

Here, s_i, t_i, u, v are **terms** in the language of $\mathbf{A}$ over the variables $\mathbf{x}$.

Asked: Does Φ hold in $\mathbf{A}$?

Computational Complexity: For finite $\mathbf{A}$ of finite type, $\text{QUASIDVAL}(\mathbf{A})$ is in **co-NP**:

$\mathbf{a} \in A^n$ witnesses failure of Φ if $\left(\bigwedge_{i \in \underline{k}} s_i^{\mathbf{A}}(\mathbf{a}) = t_i^{\mathbf{A}}(\mathbf{a}) \right) \wedge u^{\mathbf{A}}(\mathbf{a}) \neq v^{\mathbf{A}}(\mathbf{a})$.

Exponential time method: A quasi-identity of length ℓ contains at most ℓ different variables that can take at most $|A|^\ell$ values.

Question: For which algebras do we have faster methods (e.g. polynomial time)?

Quasi-identity validity and polynomial systems

Relations to other problems:

- ▶ If we can decide solvability of polynomial systems, then we can check the validity of quasi-identities.

Quasi-identity validity and polynomial systems

Relations to other problems:

- ▶ If we can decide solvability of polynomial systems, then we can check the validity of quasi-identities.
- ▶ We search for a counter-example: $\forall \mathbf{x} : \left(\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}) \right) \implies u(\mathbf{x}) = v(\mathbf{x})$ holds iff for all $a, b \in A$ with $a \neq b$,

$$\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}), \quad u(\mathbf{x}) = a, \quad v(\mathbf{x}) = b$$

has no solution.

- ▶ These systems use constants: a and b .
Therefore they are **polynomial systems** and not just **term systems**.

Quasi-identity validity and polynomial systems

Relations to other problems:

- ▶ If we can decide solvability of polynomial systems, then we can check the validity of quasi-identities.
- ▶ We search for a counter-example: $\forall \mathbf{x} : \left(\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}) \right) \implies u(\mathbf{x}) = v(\mathbf{x})$ holds iff for all $a, b \in A$ with $a \neq b$,

$$\bigwedge_{i \in \underline{k}} s_i(\mathbf{x}) = t_i(\mathbf{x}), \quad u(\mathbf{x}) = a, \quad v(\mathbf{x}) = b$$

has no solution.

- ▶ These systems use constants: a and b .
Therefore they are **polynomial systems** and not just **term systems**.
- ▶ **Conclusion:** $\text{QUASIDVAL}(\mathbf{A}) \leq_{\text{truth table}} \text{POLSYSAT}(\mathbf{A})$.

Quasi-identity validity and systems of term equations

- If we can check the validity of quasi-identities, then we can decide solvability of term equations.

Quasi-identity validity and systems of term equations

- ▶ If we can check the validity of quasi-identities, then we can decide solvability of term equations.
- ▶ The system $s_1 = t_1, \dots, s_k = t_k$ has no solution iff

$$s_1 = t_1 \wedge \dots \wedge s_k = t_k \implies y = z$$

is valid in **A**. (y, z ... new variables, $|A| > 1$).

Quasi-identity validity and systems of term equations

- ▶ If we can check the validity of quasi-identities, then we can decide solvability of term equations.
- ▶ The system $s_1 = t_1, \dots, s_k = t_k$ has no solution iff

$$s_1 = t_1 \wedge \dots \wedge s_k = t_k \implies y = z$$

is valid in $\mathbf{A}$. ($y, z \dots$ new variables, $|A| > 1$).

- ▶ **Conclusion:** $\text{co-TERMSYSSAT}(\mathbf{A}) \leq_P \text{QUASIIDVAL}(\mathbf{A})$.

Quasi-identity validity and checking term equivalence

- If we can check the validity of quasi-identities, we can check whether two terms induce the same function.

Quasi-identity validity and checking term equivalence

- ▶ If we can check the validity of quasi-identities, we can check whether two terms induce the same function.
- ▶ $\forall \mathbf{x} : s(\mathbf{x}) = t(\mathbf{x})$ is valid iff

$$y = y \implies s(\mathbf{x}) = t(\mathbf{x})$$

is valid in $\mathbf{A}$.

Quasi-identity validity and checking term equivalence

- ▶ If we can check the validity of quasi-identities, we can check whether two terms induce the same function.
- ▶ $\forall \mathbf{x} : s(\mathbf{x}) = t(\mathbf{x})$ is valid iff

$$y = y \implies s(\mathbf{x}) = t(\mathbf{x})$$

is valid in $\mathbf{A}$.

- ▶ **Conclusion:** $\text{TERM EQV}(\mathbf{A}) \leq_P \text{QUASI ID VAL}(\mathbf{A})$.

Quasi-identity validity: connections with well-studied problems.

Connections:

- ▶ $\text{QUASIDVAL}(\mathbf{A}) \leq_{\text{truth table}} \text{POLSYSSAT}(\mathbf{A})$.
- ▶ $\text{co-TERMSYSSAT}(\mathbf{A}) \leq_P \text{QUASIDVAL}(\mathbf{A})$.
- ▶ $\text{TERMEQV}(\mathbf{A}) \leq_P \text{QUASIDVAL}(\mathbf{A})$.

Quasi-identity validity: connections with well-studied problems.

Connections:

- ▶ $\text{QUASIIDVAL}(\mathbf{A}) \leq_{\text{truth table}} \text{POLSYSSAT}(\mathbf{A})$.
- ▶ $\text{co-TERMSYSSAT}(\mathbf{A}) \leq_P \text{QUASIIDVAL}(\mathbf{A})$.
- ▶ $\text{TERMEQV}(\mathbf{A}) \leq_P \text{QUASIIDVAL}(\mathbf{A})$.
- ▶ In 2004, M. Volkov constructed a 10-element semigroup $\mathbf{Q}$ with $\text{TERMEQV}(\mathbf{Q}) \in \mathbf{P}$, and $\text{QUASIIDVAL}(\mathbf{Q})$ co-**NP**-complete because it solves 3-COLORABILITY for graphs.

Quasi-identity validity: connections with well-studied problems.

Let $\mathbf{A}$ be an algebra with a Mal'cev term.

Consequences:

- ▶ $\mathbf{A}$ is abelian $\implies \text{QUASIDVAL}(\mathbf{A}) \in \mathbf{P}$.

(Reason: POLSYSAT, which is analyzed in [Larose, Zádori 2006])

Quasi-identity validity: connections with well-studied problems.

Let $\mathbf{A}$ be an algebra with a Mal'cev term.

Consequences:

- ▶ $\mathbf{A}$ is abelian $\implies \text{QUASIDVAL}(\mathbf{A}) \in \mathbf{P}$.
(Reason: POLSYSAT, which is analyzed in [Larose, Zádori 2006])
- ▶ $\text{Core}(\mathbf{A})$ is nonabelian $\implies \text{QUASIDVAL}(\mathbf{A})$ is co-**NP**-complete.
(Reason: TERMSYSAT, which is analyzed in [Mayr 2023])

Quasi-identity validity: connections with well-studied problems.

Let $\mathbf{A}$ be an algebra with a Mal'cev term.

Consequences:

- ▶ $\mathbf{A}$ is abelian $\implies \text{QUASIDVAL}(\mathbf{A}) \in \mathbf{P}$.
(Reason: POLSYSAT, which is analyzed in [Larose, Zádori 2006])
- ▶ $\text{Core}(\mathbf{A})$ is nonabelian $\implies \text{QUASIDVAL}(\mathbf{A})$ is co-**NP**-complete.
(Reason: TERMSYSAT, which is analyzed in [Mayr 2023])
- ▶ $\mathbf{A}$ non-solvable group $\implies \text{QUASIDVAL}(\mathbf{A})$ is co-**NP**-complete.
(Reason: TERMSEQV, which is analyzed in [Horváth, Lawrence, Mériai, Szabó 2007])

Quasi-identity validity: connections with well-studied problems.

Let $\mathbf{A}$ be an algebra with a Mal'cev term.

Consequences:

- ▶ $\mathbf{A}$ is abelian $\implies \text{QUASIDVAL}(\mathbf{A}) \in \mathbf{P}$.
(Reason: POLSYSAT, which is analyzed in [Larose, Zádori 2006])
- ▶ $\text{Core}(\mathbf{A})$ is nonabelian $\implies \text{QUASIDVAL}(\mathbf{A})$ is co-**NP**-complete.
(Reason: TERMSYSAT, which is analyzed in [Mayr 2023])
- ▶ $\mathbf{A}$ non-solvable group $\implies \text{QUASIDVAL}(\mathbf{A})$ is co-**NP**-complete.
(Reason: TERMSEQV, which is analyzed in [Horváth, Lawrence, Mériai, Szabó 2007])

Open: nonabelian nilpotent groups, nonzero nilpotent rings.

A reduction of graph coloring to quasi-identities

Quasi-identity validity

Theorem Aichinger, Grünbacher, STACS 2023

$\mathbf{A}$ finite algebra of finite type with a Mal'cev term. Then

1. $\text{QUASIDVAL}(\mathbf{A}) \in \mathbf{P}$ if $\mathbf{A}$ is abelian.
2. $\text{QUASIDVAL}(\mathbf{A})$ is co- $\mathbf{NP}$ -complete if $\mathbf{A}$ is nonabelian.

New content: item (2).

Proof idea: we reduce the H -coloring problem to $\text{QUASIDVAL}(\mathbf{A})$.

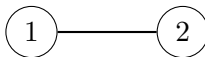
H -coloring of graphs

H -COLORING:

Given: a graph G .

Asked: Is there a graph homomorphism h from G to H ($G \rightarrow H$)?

► $H = K_2$:



$G \rightarrow H$ iff G is bipartite: edges in G only go from $h^{-1}(\{1\})$ to $h^{-1}(\{2\})$.

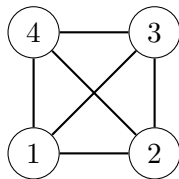
H -coloring of graphs

H -COLORING:

Given: a graph G .

Asked: Is there a graph homomorphism h from G to H ($G \rightarrow H$)?

► $H = K_4$:



$G \rightarrow H$ if the vertices of G can be coloured with 4 colors such that no adjacent vertices have the same colour.

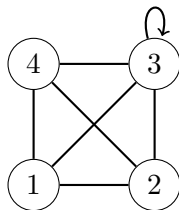
H -coloring of graphs

H -COLORING:

Given: a graph G .

Asked: Is there a graph homomorphism h from G to H ($G \rightarrow H$)?

- H a graph with loops:



$G \rightarrow H$ holds for every graph G : use $h(v) = 3$ for each vertex v of G .

Theorem Hell, Nešetřil 1990.

Let H be a finite loopless graph that contains a triangle. Then H -COLORING is **NP**-complete.

A consequence stated in CSP-language:

Theorem

Let $\mathbb{H} = (H, \rho)$ be a relational structure with an antireflexive and symmetric binary relation ρ .

If $\mathbb{H}$ has $\mathbb{K}_3 = (\{1, 2, 3\}; \neq)$ as a substructure, then $\text{CSP}(\mathbb{H})$ is **NP**-complete.

Proof of the Theorem

Plan:

- ▶ We want to prove that checking the validity of quasi-identities of $\mathbf{R} := (3\mathbb{Z}_{27}, +, -, \cdot, 0)$ is co-**NP**-complete.
- ▶ We will show: there is a graph H such that

for every graph G : $G \rightarrow H \iff$ the quasi-identity $\Phi(G)$ is not valid.

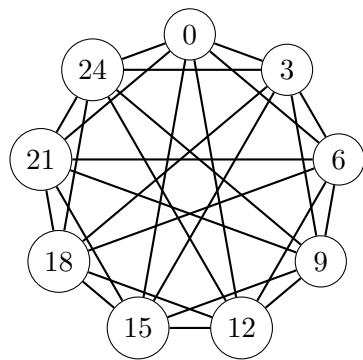
- ▶ This will imply that $\text{QUASIIDVAL}(\mathbf{R})$ is co-**NP**-complete.

Details:

- ▶ $R = \{[0]_{27}, [3]_{27}, \dots, [24]_{27}\}$.
- ▶ H is the “difference graph” or “apartness graph” on R :
 (r, s) is an edge if $r - s \notin \{[0]_{27}, [9]_{27}, [18]_{27}\}$.

Proof of the Theorem

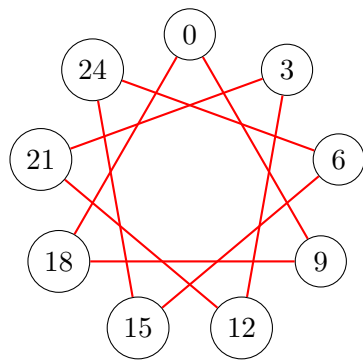
The graph H for $3\mathbb{Z}_{27}$



$$E(H) = \{(x, y) \mid x - y \notin \{0, 9, 18\}\}.$$

Proof of the Theorem

The graph H for $3\mathbb{Z}_{27}$

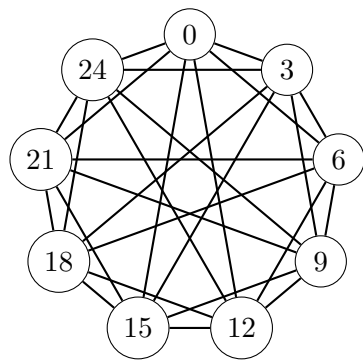


$$E(H) = \{(x, y) \mid x - y \notin \{0, 9, 18\}\}.$$

► non-edges of H

Proof of the Theorem

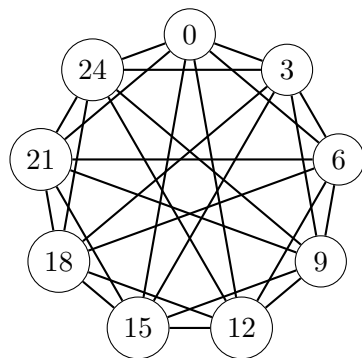
The graph H for $3\mathbb{Z}_{27}$



- ▶ G graph. We want to find out whether $G \rightarrow H$ using a quasi-identity on $\mathbf{R}$.
- ▶ $\Phi = \left(\bigwedge_{(u,v) \in E(G)} a = z_{u,v} \cdot (x_u - x_v) \right) \Rightarrow a = 0$.

Proof of the Theorem

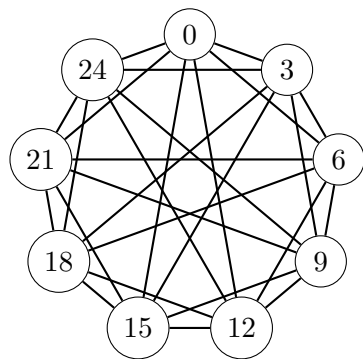
The graph H for $3\mathbb{Z}_{27}$



- ▶ G graph. We want to find out whether $G \rightarrow H$ using a quasi-identity on $\mathbf{R}$.
- ▶ $\Phi = \left(\bigwedge_{(u,v) \in E(G)} a = z_{u,v} \cdot (x_u - x_v) \right) \Rightarrow a = 0$.
- ▶ Suppose Φ is invalid. Then $a \neq 0$.
- ▶ Let $(u, v) \in E(G)$. Then $x_u - x_v \notin \{0, 9, 18\}$.
- ▶ Thus (x_u, x_v) is an edge of H .
- ▶ $u \mapsto x_u$ is a homomorphism from G to H .
- ▶ Hence if Φ is invalid, $G \rightarrow H$.

Proof of the Theorem

The graph H for $3\mathbb{Z}_{27}$



- ▶ G graph. We want to find out whether $G \rightarrow H$ using a quasi-identity on $\mathbf{R}$.
- ▶ $\Phi = \left(\bigwedge_{(u,v) \in E(G)} a = z_{u,v} \cdot (x_u - x_v) \right) \Rightarrow a = 0$.
- ▶ Suppose $G \rightarrow H$.
- ▶ ...
- ▶ This is a counterexample to Φ .
- ▶ Hence Φ is invalid.

Proof of the Theorem

- ▶ Hence Φ is not valid iff $G \rightarrow H$.
- ▶ H -coloring is **NP**-complete [Hell, Nešetřil 1990].
- ▶ Thus $\text{QUASIDVAL}(\mathbf{R})$ is co-**NP**-complete.

Proof for Mal'cev algebras

Theorem

Let $\mathbf{A}$ be a finite nonabelian algebra of finite type with a Mal'cev term. Then $\text{QUASIDVAL}(\mathbf{A})$ is co-**NP**-complete.

- ▶ Instead of the ring multiplication, use commutators [Smith 1976, Hagemann, Herrmann 1979].
- ▶ This works for subdirectly irreducible $\mathbf{A}$.
- ▶ For arbitrary $\mathbf{A}$, use “difference graphs” for several congruences of $\mathbf{A}$.
- ▶ Order these graphs and pick a maximal one.
- ▶ EA and Simon Grünbacher. The Complexity of Checking Quasi-Identities over Finite Algebras with a Mal'cev Term, STACS 2023.

IV. Open questions

Polynomial equations on S_4

Problem: Solve $\text{POLSAT}(S_4)$ or $\text{POLEQV}(S_4)$ in time $m^{c \log^d(m)}$.

Possible approaches:

- ▶ Show: every nonconstant absorbing polynomial in k arguments has length at least $2^{ck^{1/d}}$.
- ▶ The following problems are – in some sense – equivalent:
 - ▶ Does a fully expanded polynomial $p \in \text{Mat}_2(\mathbb{F}_2) \langle X_1, \dots, X_n \rangle$ vanish on all 6^n invertible inputs?
 - ▶ (Grünbacher) Let ω be primitive in $\mathbb{F}_4$, $p_1, \dots, p_n$ expanded polynomials in $\mathbb{Z}_3[X_1, \dots, X_n]$. Does

$$\sum_{i=1}^n \omega^{p_i(X_1, \dots, X_n)}$$

vanish on $\{-1, 1\}^n$?

Circuit equivalence

- ▶ The polynomials p, q may be given as circuits.
- ▶ We want to test whether they compute the same function.

Problem 1 from [Kawałek, Kompatscher, Krzaczkowski, STACS 2024]

Let $\mathbf{A}$ be a finite algebra from a congruence modular variety with supernilpotent rank 2. Is there a deterministic polynomial time algorithm solving circuit equivalence on $\mathbf{A}$?

Děkuji za pozvání a pozornost!